

Aaron Lin

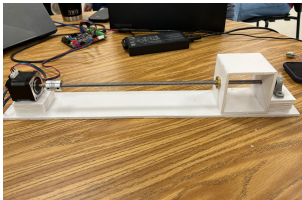
Rocket hardware, autonomous prototypes, and testing

B.S. Mechanical Engineering, Aerospace Concentration · Boston University · May 2028

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I am a mechanical engineering junior seeking a **Summer 2027 internship** in mechanical design or aerospace. My recent work spans electromechanical motion, supersonic fairing development, and rocket hardware from analysis through fabrication and test integration.

Selected engineering work



2.5-DOF Cartesian motion system

Team project · In progress

Lead-screw motion stages, motorized pen lift, G-code control, and early square and circle path testing.

Paths tested

coordinated X-Y motion and pen actuation demonstrated



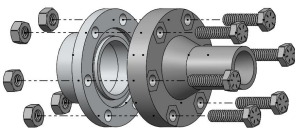
ARGO aerodynamic fairing

My role: project engineer · In progress

Installed-geometry fairing concepts for an external downcomer and wiring at a Mach 2.81 design condition.

CDR complete

September 2026; comparative analysis in progress



Converging-diverging rocket nozzle

My role: individual mechanical lead · Completed

Flow sizing, CAD, drawings, pressure analysis, resin printing, and static-fire integration.

2.12

minimum factor of safety in the pressure-only model

Experience and capabilities

Mechanical Engineering Intern, Culturon

Sydney, Australia · February–June 2026

Ran optical emission spectroscopy experiments and built Python tools with pandas, SciPy, and astroquery to process plasma spectra and compare candidate lines with the NIST Atomic Spectra Database.

Project Engineer, BU Rocket Propulsion Group

September 2024–present

Designed and fabricated a resin rocket nozzle; now comparing aerodynamic fairing concepts using Ansys Fluent, OpenRocket, and first-principles calculations.

Design and analysis

SolidWorks · Ansys Fluent · OpenRocket · FEA · CFD · GD&T fundamentals

Programming and data

Python · pandas · SciPy · astroquery · MATLAB · Arduino C++

Fabrication and test

Resin and FDM printing · CNC and manual machining · soldering · sensors · experimental testing

More drawings, reports, test media, and complete case studies: aaronlin.space.

2.5-DOF Cartesian motion system

Boston University ME360 · Team project · September 2026–present · In progress

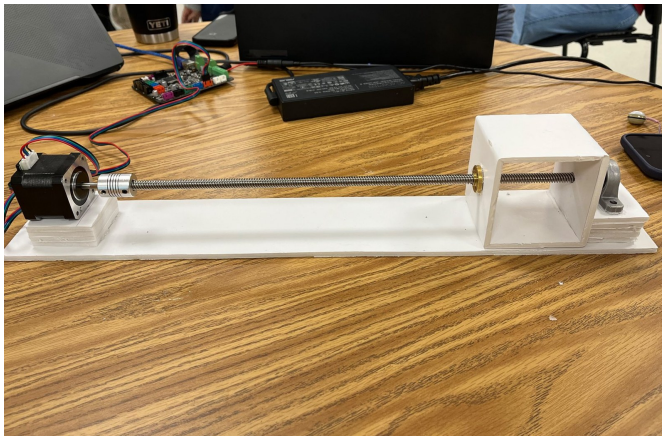
My role: Team member **My contribution:** Individual contribution is not yet documented in the current project record

Team outcome: Early prototype completed coordinated square and circle path tests **Evidence status:** Motion demonstrated; positioning accuracy and repeatability unmeasured

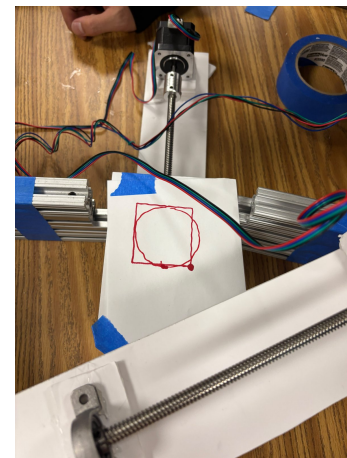
Our team is building a pen plotter that converts G-code into coordinated X-Y motion and a motorized pen lift. The early prototype uses perpendicular lead-screw stages, stepper motors, and temporary foam-board structures to expose motion and control problems before the final structure is built.

Square + circle

recognizable paths produced in coordinated motion tests



Early lead-screw axis used to test the motion architecture.



First combined path test; rounded corners and line waviness remain visible.

What the current prototype shows

The test hardware demonstrated coordinated motion across two perpendicular axes and actuated the pen during commanded paths. The square and circle are recognizable, which makes the prototype useful for finding control and structural issues before the final build.

Problems exposed by testing

Rounded corners and line waviness remain in the traces. The next iteration needs a stiffer frame, better-constrained carriages, calibrated travel per motor step, and repeated measurements across the required work area.

Ownership note: This page reports the team's prototype evidence. My specific design and implementation contribution will be added after it is documented in the final project record.

ARGO aerodynamic fairing

BU Rocket Propulsion Group · Project Engineer · July 2026–present · In progress

My role: Project engineer **My contribution:** Baseline and candidate geometry, stagnation calculations, material screening, and CFD preparation

Team outcome: Conceptual Design Review completed **Evidence status:** Current milestone documented; comparative simulation results pending



Installed packaging model showing ARGO's external downcomer and an early aft-fairing concept.

Design problem

ARGO routes a downcomer and wiring along the rocket exterior, creating aerodynamic, thermal, attachment, and packaging challenges. The fairing must protect those components while limiting drag and remaining compatible with the installed geometry.

My work

I am comparing the previous Icarus aerocover with candidate leading edges, midsections, and trailing-edge or boattail concepts using Ansys Fluent, OpenRocket, and first-principles calculations. I translate the installed packaging constraints into consistent baseline and candidate geometries and document the assumptions used in each comparison.

Completed milestone

The team completed Conceptual Design Review on September 3, 2026. At the **Mach 2.81 design condition**, work to date includes stagnation temperature and pressure calculations, early material screening, and preparation of candidate CFD geometry.

Comparison controls

A fairing attached to the rocket body is only a partial revolution, so results from a fully revolved nose cone do not transfer directly. Baseline and candidate concepts must use the same installed geometry, boundary conditions, reference area, and reporting method before their aerodynamic results can be compared.

Limits: This is an in-progress design review, not a completed performance claim. No drag reduction, pressure result, or selected geometry is reported until the baseline-versus-candidate simulations are complete and checked for mesh and setup consistency.

Next analysis

Before Preliminary Design Review, I will run the baseline and candidate with matched geometry, boundary conditions, reference area, and reporting methods. I will document the mesh and compare one checked performance metric.

Methods: Ansys Fluent · OpenRocket · supersonic flow · first-principles calculations

[Open the current ARGO case study](#)

Converging-diverging rocket nozzle

BU Rocket Propulsion Group · Fall 2024 · Completed

My role: Individual mechanical lead **My contribution:** Flow sizing, CAD, drawings, pressure analysis, fabrication, and integration

Team outcome: Integrated with the chamber and ignition system for static-fire testing **Evidence status:** Pressure model and build artifacts documented; performance not measured

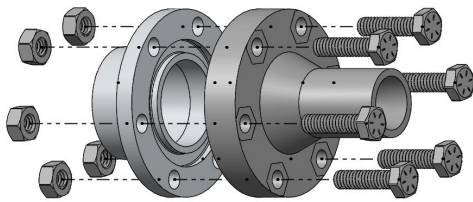
I owned the nozzle's mechanical design and fabrication. I sized the flow path with one-dimensional compressible-flow relations, produced the CAD and drawings, printed the parts, and integrated the assembly for static-fire testing. The design used an 80 psi chamber-pressure requirement, 3250 K chamber temperature, 13 g/s mass flow, and a 2.1 psia target exit pressure.

6.521

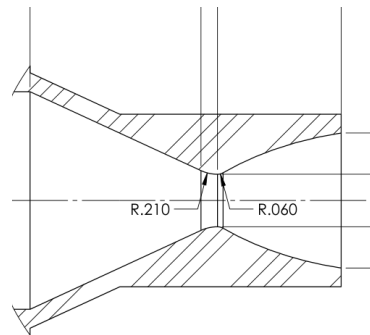
exit-to-throat area ratio

2.12

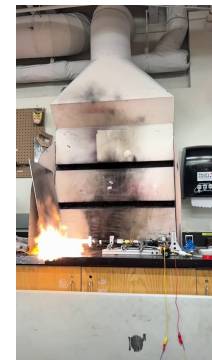
minimum factor of safety; pressure-only model



Nozzle, six-bolt adapter, and fasteners.



Internal flow path and controlling dimensions.



Static-fire integration recording.

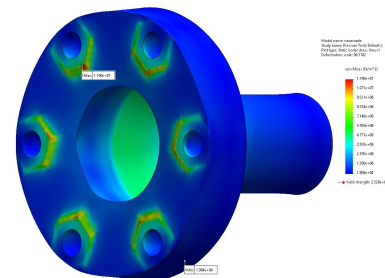
Design decisions

I used isentropic-flow relations to size the throat and exit, kept the assembly within a 1.05 in diameter limit, and modeled the nozzle and adapter in SolidWorks. I selected resin printing for the internal geometry and schedule, produced the drawings, fabricated the nozzle, and integrated it with the chamber and seals.

Pressure-analysis evidence

The SolidWorks Simulation model applied approximately 46.3 psi at the throat and 80 psi in the chamber. It reported a maximum von Mises stress of 11.90 MPa against an assumed 25.20 MPa yield strength.

Limits: The 2.12 factor of safety applies only to the pressure model and its material and boundary assumptions. The hot-fire recording demonstrates system integration, not quantified nozzle performance.



11.90 MPa maximum stress versus 25.20 MPa assumed yield; deformation exaggerated.

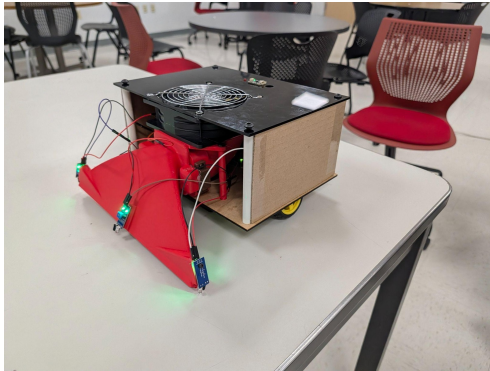
Next validation. Add temperature-dependent properties, heat transfer, mesh convergence, pressure and wall-temperature measurements, and a documented post-fire inspection.

Methods: SolidWorks · FEA · one-dimensional compressible flow · resin printing

[Open the nozzle case study](#)

Selected build, test, and analysis work

Earlier projects and industry experience · Supporting evidence



Prototype used for presentation and pickup testing.

The Deskinator

Boston University EK210 · Team of four · Fall 2025

I wrote the Arduino C++ navigation firmware for S-curve pathing, IR-triggered edge avoidance, and gesture activation, and helped assemble and troubleshoot the mechanical and electrical systems.

76% pickup rate in one documented final trial: 72 of 95 rice grains

The team improved pickup after changing the drive motors, fan, filter, and nozzle. Edge avoidance and gesture activation were demonstrated but not fully repeatable.

[Open the full Deskinator case study](#)

Mechanical Engineering Intern, Culturon

Sydney, Australia · February–June 2026

I ran optical emission spectroscopy experiments and built Python tools with pandas, SciPy, and astroquery to process plasma spectra and compare candidate lines with the NIST Atomic Spectra Database.

This experience combined experimental setup, data processing, and source-backed interpretation of plasma spectra.

Structural mechanics laboratory studies

University of Sydney · Spring 2026 · Shared measurements

At the largest applied load, the strain gauge measured **71.44 microstrain** and the curvature calculation predicted **72.63 microstrain**, a **1.63%** difference. The beam model predicted measured deflections within about 10% across the load steps.

Authorship note: I used measurements gathered with classmates in my calculations and reports; these values are presented as team evidence.

Additional work online

[Temperature monitor](#)

Individual Arduino build with sensing, dual-unit display, and high- and low-temperature alerts.

[Diffusion analysis](#)

Sample A produced a 23.6 kJ/mol activation-energy fit; sample B remained inconclusive.

[Acrylic truss](#)

The final MATLAB model predicted 46.59 oz; the physical build met the 16 oz minimum.

[SolidWorks exercises](#)

Parts, assemblies, section views, exploded views, drawings, bills of materials, and motion studies.

See the full case studies, drawings, reports, and test media at aaronlin.space.

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